Stance-phase kinematic phenotyping is associated with patient-reported function in moderate knee osteoarthritis

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Introduction

Knee osteoarthritis (KOA) is a leading cause of disability in older women. Yet, moderate disease (Kellgren–Lawrence grades 2–3) is clinically heterogeneous: patients with matched radiographic severity diverge into conservative or surgical pathways while patient-reported outcomes (WOMAC, HADS) overlap substantially across groups [1]. Radiographic and symptomatic descriptors, however, miss biomechanical information relevant to stratification. To fill this gap, we tested whether unsupervised phenotyping of stance-phase kinematics identifies patterns that align more closely with patient-reported function than with hospital treatment decision.

Methods

66 women aged 60–75 with moderate KOA - 27 total knee replacement-scheduled (TKR), 39 conservatively managed (CONS) - underwent gait analysis. The difference of the value of the angle at terminal stance (t_3) and loading phase (t_1) was computed as Δ = Angle(t_3)-Angle(t_1) for twelve kinematic variables. Z-scored features were clustered with k-means (k=2, selected by convergence of gap statistic, silhouette, Davies–Bouldin, and Ward dendrogram). Stability used Hennig bootstrap (B=1000, Jaccard ≥ 0.85) [2]. Clusters were compared across the 12 Δ variables and 5 primary clinical outcomes (WOMAC pain /function /stiffness; HADS anxiety /depression) with Welch t or Mann–Whitney tests, Hedges' g, and Benjamini–Hochberg FDR within each family. Logistic regression compared clinical scales against the binary treatment hospital decision (TKR, CONS) as predictors of cluster membership.

Results

Two highly stable clusters emerged (bootstrap Jaccard 0.95 and 0.92; ARI k-means vs Ward = 0.86). Nine of twelve Δ variables differed between clusters after FDR correction (Fig. 1); shoulder rotation offset (g = +4.65), back flexion–extension (g = –3.65) and pelvic tilt (g = +3.17) contributed the most. Cluster 0 (n = 52) increased anterior pelvic tilt across stance, while Cluster 1 (n = 14) decreased it. WOMAC pain, function, and stiffness were higher in Cluster 0 after FDR; HADS subscales were not. Cluster 1 fell in low WOMAC tertiles (86% function, 79% pain), with 17–21% cluster–tertile discordance (Fig. 2A). Cluster membership was best predicted by WOMAC (AUC = 0.92, McFadden R^2 = 0.44), followed by treatment classification (AUC = 0.71, R^2 = 0.15), while HADS did not discriminate above chance (AUC = 0.59, R^2 = 0.02) (Fig. 2B).

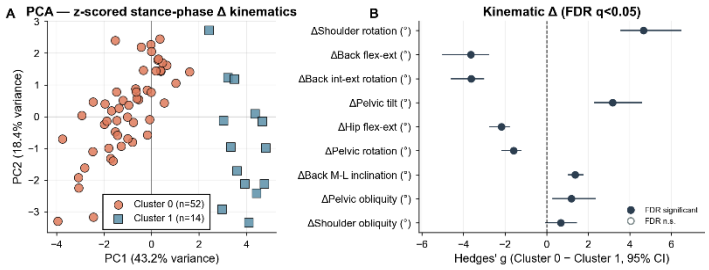


Figure 1: (A) Principal component projection of z-scored stance-phase Δ kinematics colored by k-means cluster (k=2). (B) Kinematic Δ variables surviving FDR correction; Hedges' g with 95% CI.

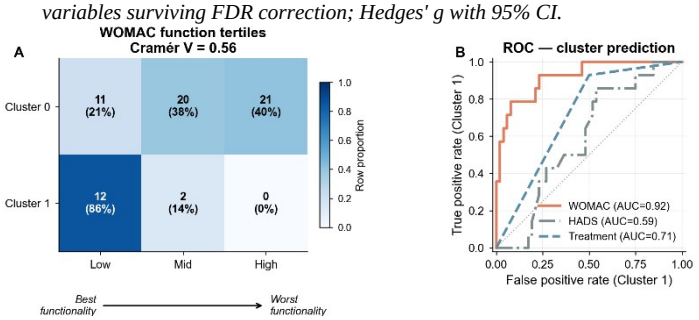


Figure 2: (A) WOMAC function tertile distribution across kinematic clusters (Cramér V = 0.56), Cluster 1 concentrates in the low-severity tertile. (B) ROC curves for logistic models predicting cluster 1 membership: WOMAC (AUC = 0.92) outperforms both HADS (0.59) and the binary treatment classification (0.71).

Discussion and Conclusions

Stance-phase kinematics separate women with moderate KOA into two patterns that align more closely with patient-reported function than with the surgical-versus-conservative pathway. Cluster membership was strongly but not fully explained by WOMAC, with 17–21% cluster–tertile discordance, suggesting that kinematics carries partially non-redundant information. Interestingly, Cluster 1 contains the most functional patients, a phenotype invisible to KL grading and only partially captured by WOMAC. Given the small size of Cluster 1 (n = 14), effect-size and discrimination estimates require confirmation in larger independent samples. These findings encourage further exploration of motion descriptors in stratification frameworks for moderate KOA; longitudinal replication in larger cohorts is required to establish prognostic value.

Acknowledgements

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References

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